

Cor Borde Bola Pseudohiperbolica

Desigualdades Auxiliares

Corolario 1. Sean $z_1, z_2 \in \mathbb{D}$, entonces

$$(1) \frac{z_1 - z_2}{1 - \bar{z}_1 z_2} \in \partial B_\rho(z_1, |z_2|) \quad (2) \frac{|z_1| - |z_2|}{1 - |z_1 z_2|} \leq \left| \frac{z_1 - z_2}{1 - \bar{z}_1 z_2} \right| \leq \frac{|z_1| + |z_2|}{1 + |z_1 z_2|}.$$

Demostración:

(1) Basta ver que $\rho(\tau_{z_1}(z_2), z_1) = |z_2|$. Como $\tau_{z_1} \in \text{Aut}(\mathbb{D})$, por el teorema de Schwarz-Pick, τ_{z_1} es una isometría para la métrica pseudohiperbólica. Por tanto, $\rho(\tau_{z_1}(z_2), z_1) = \rho(\tau_{z_1}(z_2), \tau_{z_1}(0)) = \rho(z_2, 0) = |z_2|$.

(2) Se obtiene aplicando el `lem-cota-modulo-bola-pseudohiperbolica/lema 1` con $z = z_1$ y $r = |z_2|$ y usando ??.

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Referenciado en

- Teo Metrica Pseudohiperbolica Desigualdad Triangular Generalizada

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